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(54) **GROUP VOLUME CONTROL**

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(21) Appl. No.: **19/088,874**

(57)

ABSTRACT

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Related U.S. Application Data

(63) Continuation of application No. 18/309,577, filed on Apr. 28, 2023, now Pat. No. 12,260,064, which is a continuation of application No. 16/827,469, filed on Mar. 23, 2020, now Pat. No. 11,640,225, which is a continuation of application No. 14/205,347, filed on Mar. 11, 2014, now Pat. No. 10,599,287.

Publication Classification

(51) **Int. Cl.**

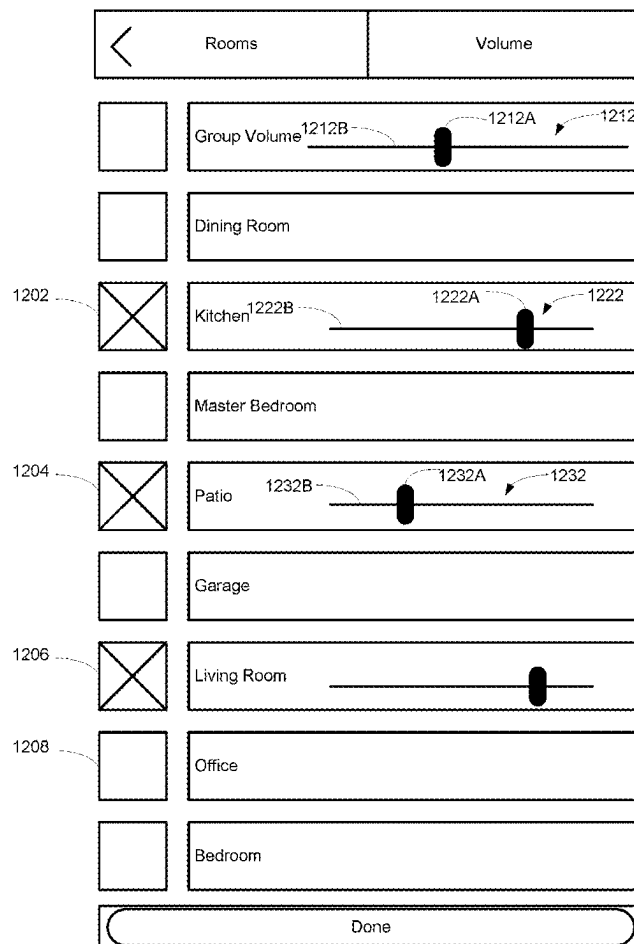
G06F 3/04817 (2022.01)

G06F 3/04847 (2022.01)

G06F 3/16 (2006.01)

H04L 67/025 (2022.01)

An example method involves providing an interface that simultaneously displays an indication of a respective volume corresponding to each of at least two zone players and displays a respective selectable icon corresponding to each of the at least two zone players. Each respective selectable icon indicates whether the corresponding zone player has been selected for a group, and at least one of the respective selectable icons indicates that a respective zone player has been selected for the group. The interface further displays an indication of a group volume associated with the at least one selected zone player. The method further includes receiving a command to change at least one of a respective volume of one or more of the at least one selected zone player or the group volume, and transmitting an indication of the received command to one or more of the at least one selected zone player.



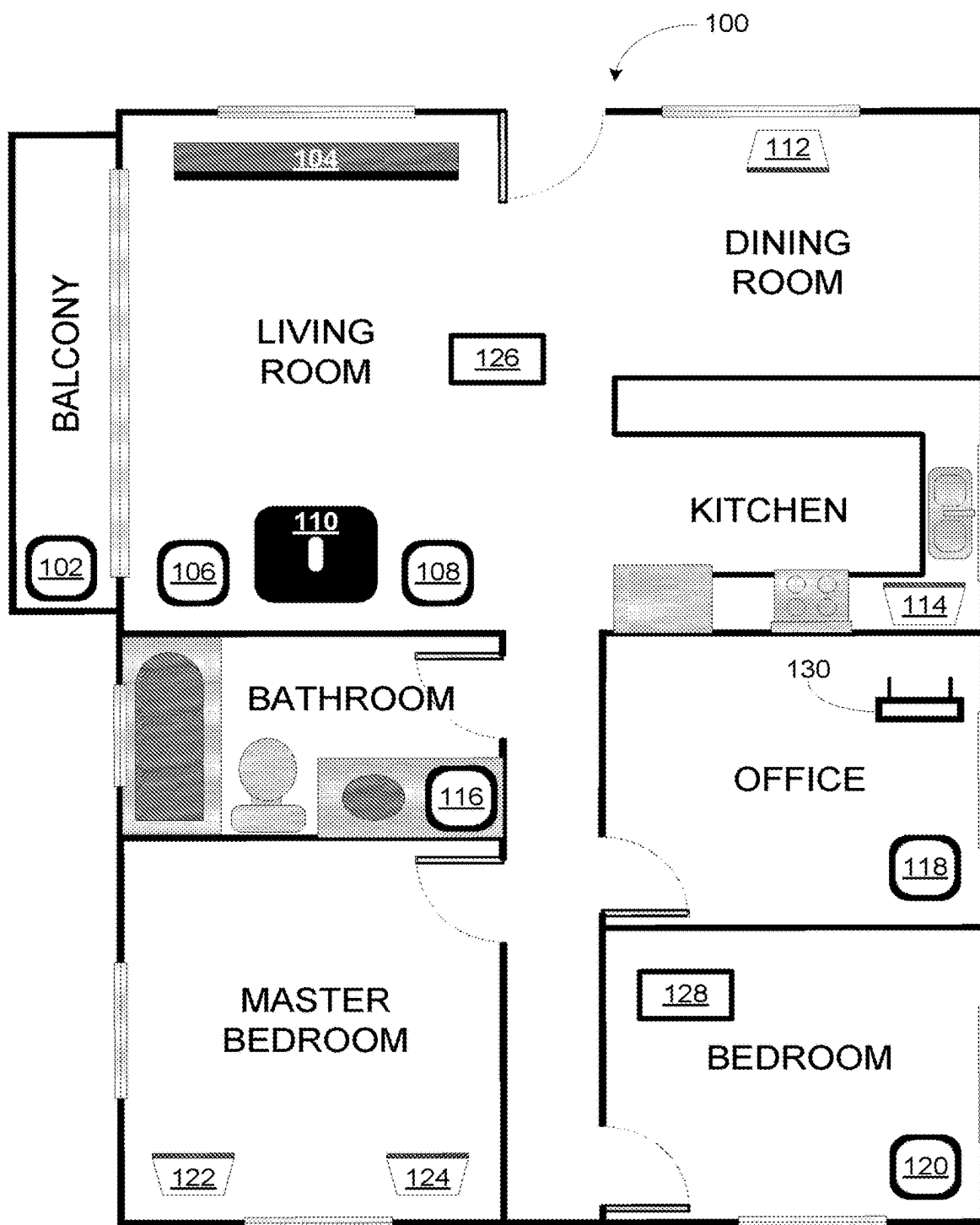


FIGURE 1

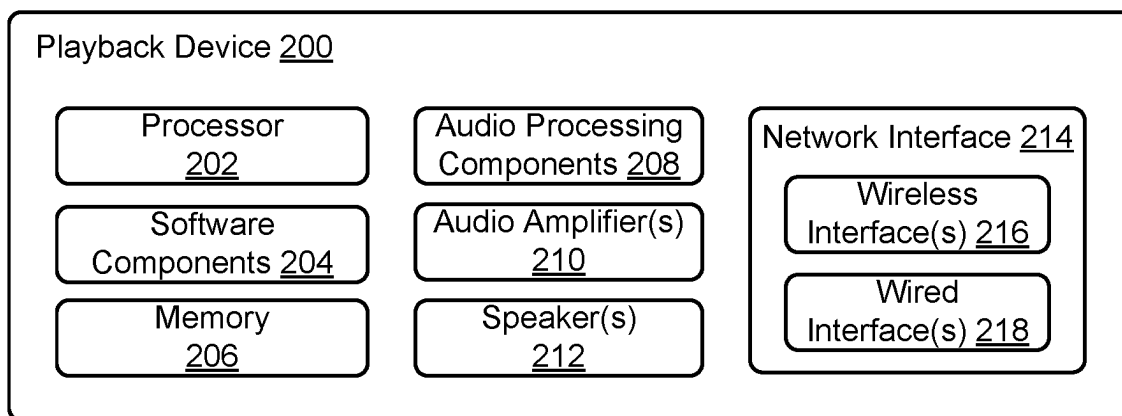


FIGURE 2

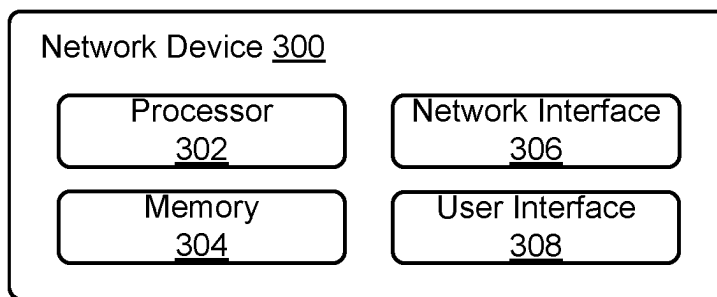


FIGURE 3

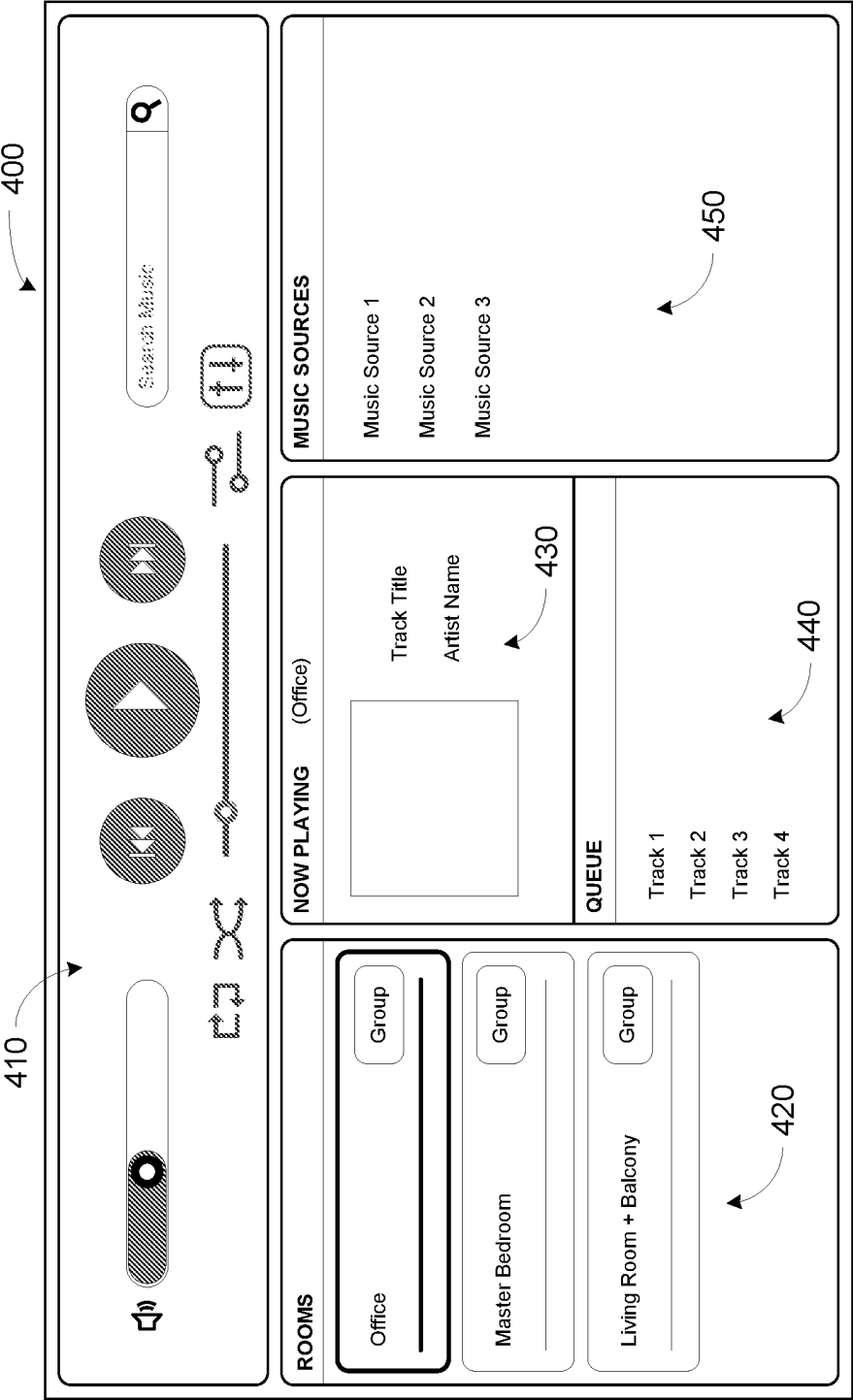


FIGURE 4

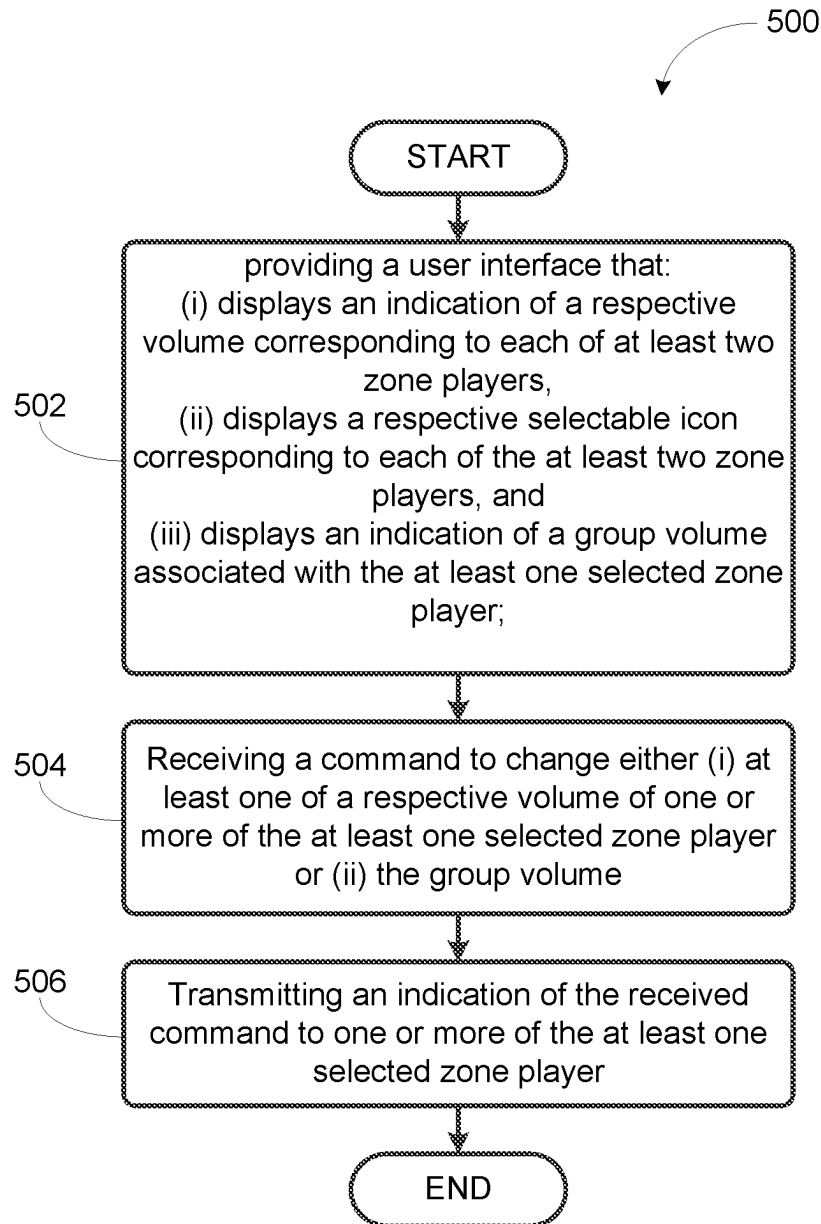


FIGURE 5

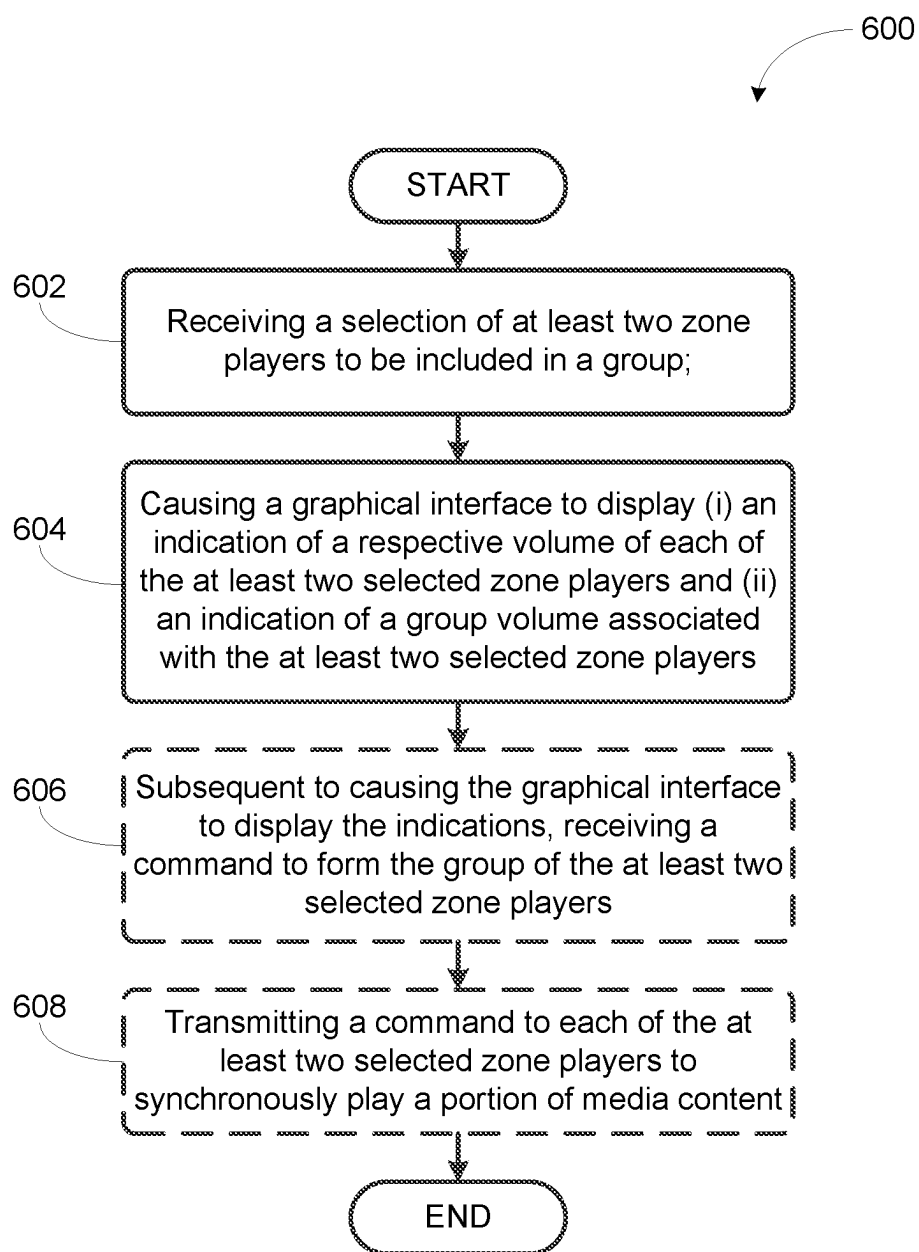


FIGURE 6

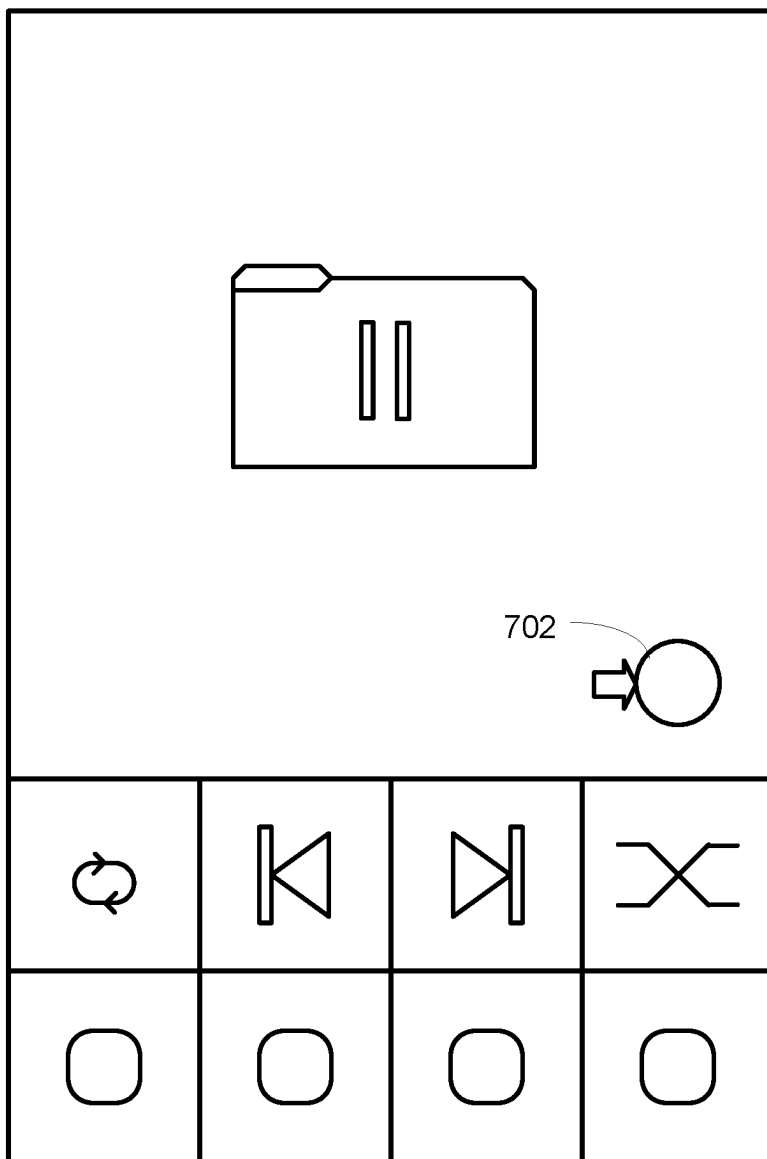


FIGURE 7

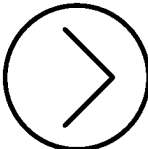
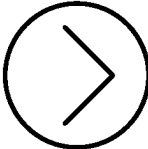
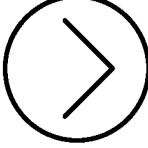
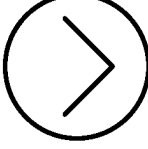
Living Room	
Dining Room Kitchen Master Bedroom Guest Bedroom	
Patio Foyer	
Entrance	
Done	

FIGURE 8

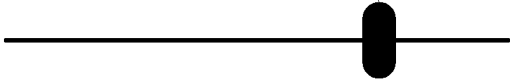
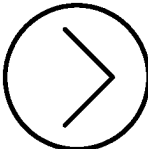
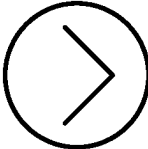
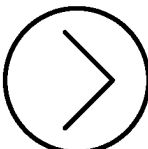
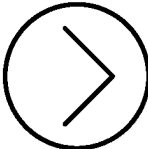
Living Room 902 	
Dining Room Kitchen Master Bedroom Guest Bedroom	
Patio Foyer	
Entrance	
Done	

FIGURE 9

<

Rooms

Volume

Group Volume

1002

☐

Dining Room

1004

☒

Kitchen

☐

Master Bedroom

☒

Patio

☐

Garage

☒

Living Room

☐

Office

☐

Bedroom

Done

FIGURE 10

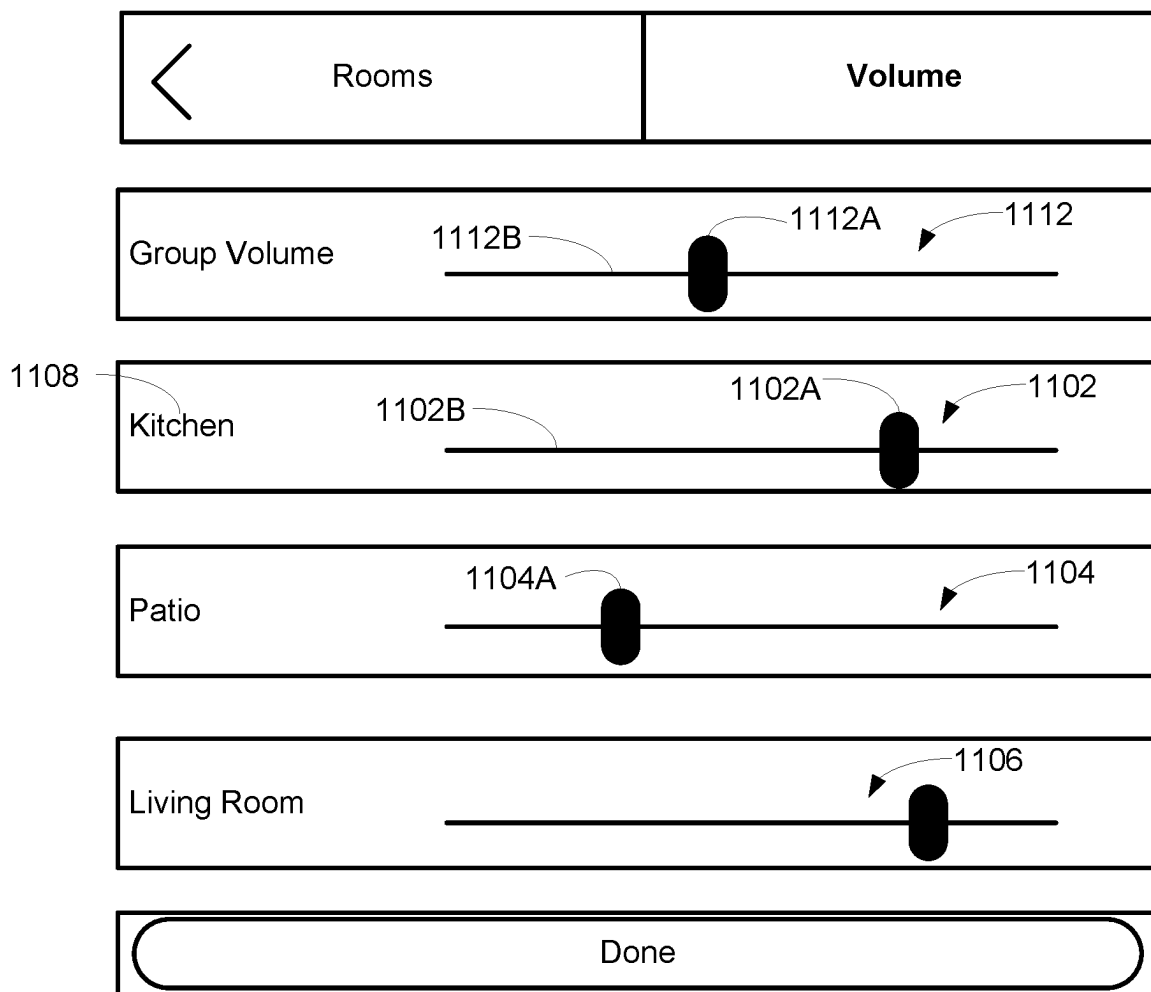


FIGURE 11

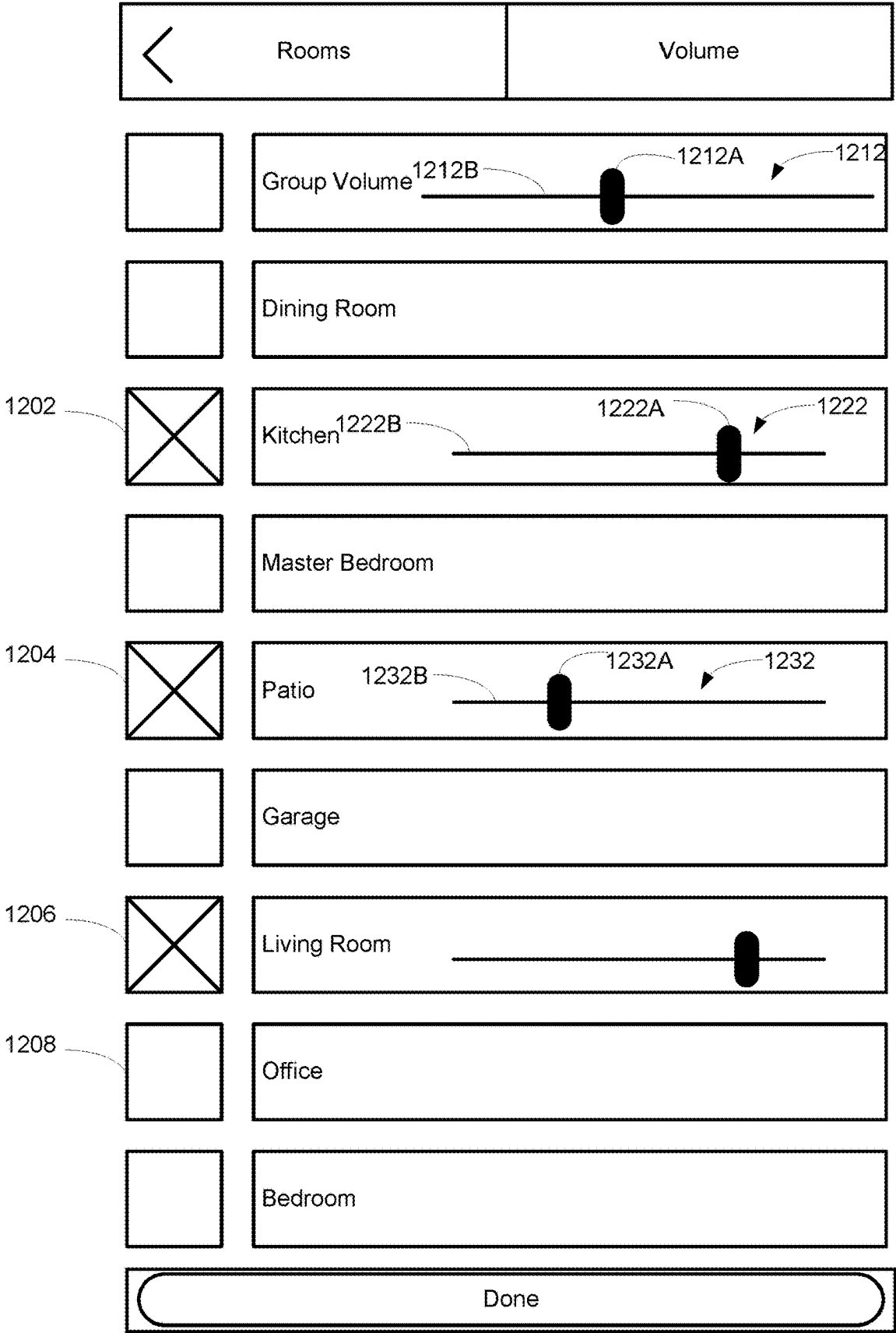


FIGURE 12

GROUP VOLUME CONTROL

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/309,577, titled “Group Volume Control,” filed on Apr. 28, 2023, and currently pending; U.S. application Ser. No. 18/309,577 is a continuation of U.S. application Ser. No. 16/827,469, titled “Group Volume Control,” filed on Mar. 23, 2020 and issued as U.S. Pat. No. 11,640,225 on Apr. 12, 2023; U.S. application Ser. No. 16/827,469 is a continuation of U.S. application Ser. No. 14/205,347, titled “Group Volume Control,” filed on Mar. 11, 2014, and issued as U.S. Pat. No. 10,599,287 on Mar. 24, 2020. The entire contents of U.S. application Ser. Nos. 18/309,577, 16/827,469, and 14/205,347 are incorporated by reference herein.

FIELD OF THE DISCLOSURE

[0002] The disclosure is related to consumer goods and, more particularly, to methods, systems, products, features, services, and other elements directed to media playback or some aspect thereof.

BACKGROUND

[0003] Options for accessing and listening to digital audio in an out-loud setting were limited until in 2003, when SONOS, Inc. filed for one of its first patent applications, entitled “Method for Synchronizing Audio Playback between Multiple Networked Devices,” and began offering a media playback system for sale in 2005. The Sonos Wireless HiFi System enables people to experience music from many sources via one or more networked playback devices. Through a software control application installed on a smartphone, tablet, or computer, one can play what he or she wants in any room that has a networked playback device. Additionally, using the controller, for example, different songs can be streamed to each room with a playback device, rooms can be grouped together for synchronous playback, or the same song can be heard in all rooms synchronously.

[0004] Given the ever growing interest in digital media, there continues to be a need to develop consumer-accessible technologies to further enhance the listening experience.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Features, aspects, and advantages of the presently disclosed technology may be better understood with regard to the following description, appended claims, and accompanying drawings where:

[0006] FIG. 1 shows an example media playback system configuration in which certain embodiments may be practiced;

[0007] FIG. 2 shows a functional block diagram of an example playback device;

[0008] FIG. 3 shows a functional block diagram of an example control device;

[0009] FIG. 4 shows an example controller interface;

[0010] FIG. 5 shows an example flow diagram for an example method.

[0011] FIG. 6 shows an example flow diagram for an example method.

[0012] FIGS. 7, 8, 9, 10, 11, and 12 show example displays.

[0013] The drawings are for the purpose of illustrating example embodiments, but it is understood that the inventions are not limited to the arrangements and instrumentality shown in the drawings.

DETAILED DESCRIPTION

I. Overview

[0014] Embodiments described herein involve, among other things, providing access to and control of a media playback network via a media playback application. The embodiments further involve providing the capability to form playback groups of zone players, from within the media playback application or from a user interface that is accessible from the media playback application. And the embodiments further involve providing for the control of playback volumes associated with the zone players, from within the media playback application or from a user interface accessible from the media playback application. In some embodiments, such control of playback volumes of zone players within a playback group may be controlled before the playback group is formed. Other aspects of the embodiments will be made apparent in the remainder of description herein.

[0015] In one aspect, a method is provided. The method involves providing a user interface that simultaneously (i) displays an indication of a respective volume corresponding to each of at least two zone players and (ii) displays a respective selectable icon corresponding to each of the at least two zone players. Each respective selectable icon indicates whether the corresponding zone player has been selected for a group, and at least one of the respective selectable icons indicates that a respective zone player has been selected for the group. The interface further simultaneously displays (iii) an indication of a group volume associated with the at least one selected zone player. The method further includes receiving a command to change either at least one of (i) a respective volume of one or more of the at least one selected zone player or (ii) the group volume, and transmitting an indication of the received command to one or more of the at least one selected zone player.

[0016] In another aspect, a device is provided. The device includes a processor and memory with instructions stored therein, that when executed cause the device to perform functions. The functions comprise providing a user interface that simultaneously (i) displays an indication of a respective volume corresponding to each of at least two zone players and (ii) displays a respective selectable icon corresponding to each of the at least two zone players. Each respective selectable icon indicates whether the corresponding zone player has been selected for a group, and at least one of the respective selectable icons indicates that a respective zone player has been selected for the group. The interface further simultaneously displays (iii) an indication of a group volume associated with the at least one selected zone player. The functions further comprise receiving a command to change at least one of (i) a respective volume of one or more of the at least one selected zone player or (ii) the group volume. The functions further comprise transmitting an indication of the received command to one or more of the at least one selected zone player.

[0017] In yet another aspect, a non-transitory computer readable memory is provided. The non-transitory computer

readable memory has stored thereon instructions executable by a computing device to cause the computing device to perform functions. The functions comprise providing a user interface that simultaneously (i) displays an indication of a respective volume corresponding to each of at least two zone players and (ii) displays a respective selectable icon corresponding to each of the at least two zone players. Each respective selectable icon indicates whether the corresponding zone player has been selected for a group, and at least one of the respective selectable icons indicates that a respective zone player has been selected for the group. The interface further simultaneously displays (iii) an indication of a group volume associated with the at least one selected zone player. The functions further comprise receiving a command to change at least one of (i) a respective volume of one or more of the at least one selected zone player or (ii) the group volume. The functions further comprise transmitting an indication of the received command to one or more of the at least one selected zone player.

[0018] In yet another aspect, a method is provided. The method involves receiving a selection of at least two zone players to be included in a group and causing a graphical interface to display (i) an indication of a respective volume of each of the at least two selected zone players and (ii) an indication of a group volume associated with the at least two selected zone players. The method further involves, subsequent to causing the graphical interface to display the indications, receiving a command to form the group of the at least two selected zone players; and transmitting a command to each of the at least two selected zone players to synchronously play a portion of media content.

[0019] In another aspect, a device is provided. The device includes a processor and memory with instructions stored therein, that when executed cause the device to perform functions. The functions comprise receiving a selection of at least two zone players to be included in a group and causing a graphical interface to display (i) an indication of a respective volume of each of the at least two selected zone players and (ii) an indication of a group volume associated with the at least two selected zone players. The functions comprise, subsequent to causing the graphical interface to display the indications, receiving a command to form the group of the at least two selected zone players; and transmitting a command to each of the at least two selected zone players to synchronously play a portion of media content.

[0020] In yet another aspect, a non-transitory computer readable memory is provided. The non-transitory computer readable memory has stored thereon instructions executable by a computing device to cause the computing device to perform functions. The functions comprise receiving a selection of at least two zone players to be included in a group and causing a graphical interface to display (i) an indication of a respective volume of each of the at least two selected zone players and (ii) an indication of a group volume associated with the at least two selected zone players. The functions further comprise, subsequent to causing the graphical interface to display the indications, receiving a command to form the group of the at least two selected zone players; and transmitting a command to each of the at least two selected zone players to synchronously play a portion of media content.

[0021] It will be understood by one of ordinary skill in the art that this disclosure includes numerous other embodiments.

II. Example Operating Environment

[0022] FIG. 1 shows an example configuration of a media playback system 100 in which one or more embodiments disclosed herein may be practiced or implemented. The media playback system 100 as shown is associated with an example home environment having several rooms and spaces, such as for example, a master bedroom, an office, a dining room, and a living room. As shown in the example of FIG. 1, the media playback system 100 includes playback devices 102-124, control devices 126 and 128, and a wired or wireless network router 130.

[0023] Further discussions relating to the different components of the example media playback system 100 and how the different components may interact to provide a user with a media experience may be found in the following sections. While discussions herein may generally refer to the example media playback system 100, technologies described herein are not limited to applications within, among other things, the home environment as shown in FIG. 1. For instance, the technologies described herein may be useful in environments where multi-zone audio may be desired, such as, for example, a commercial setting like a restaurant, mall or airport, a vehicle like a sports utility vehicle (SUV), bus or car, a ship or boat, an airplane, and so on.

a. Example Playback Devices

[0024] FIG. 2 shows a functional block diagram of an example playback device 200 that may be configured to be one or more of the playback devices 102-124 of the media playback system 100 of FIG. 1. The playback device 200 may include a processor 202, software components 204, memory 206, audio processing components 208, audio amplifier(s) 210, speaker(s) 212, and a network interface 214 including wireless interface(s) 216 and wired interface(s) 218. In one case, the playback device 200 may not include the speaker(s) 212, but rather a speaker interface for connecting the playback device 200 to external speakers. In another case, the playback device 200 may include neither the speaker(s) 212 nor the audio amplifier(s) 210, but rather an audio interface for connecting the playback device 200 to an external audio amplifier or audio-visual receiver.

[0025] In one example, the processor 202 may be a clock-driven computing component configured to process input data according to instructions stored in the memory 206. The memory 206 may be a tangible computer-readable medium configured to store instructions executable by the processor 202. For instance, the memory 206 may be data storage that can be loaded with one or more of the software components 204 executable by the processor 202 to achieve certain functions. In one example, the functions may involve the playback device 200 retrieving audio data from an audio source or another playback device. In another example, the functions may involve the playback device 200 sending audio data to another device or playback device on a network. In yet another example, the functions may involve pairing of the playback device 200 with one or more playback devices to create a multi-channel audio environment.

[0026] Certain functions may involve the playback device 200 synchronizing playback of audio content with one or more other playback devices. During synchronous playback, a listener will preferably not be able to perceive time-delay differences between playback of the audio content by the playback device 200 and the one or more other playback devices. U.S. Pat. No. 8,234,395 entitled, "System and

method for synchronizing operations among a plurality of independently clocked digital data processing devices,” which is hereby incorporated by reference, provides in more detail some examples for audio playback synchronization among playback devices.

[0027] The memory 206 may further be configured to store data associated with the playback device 200, such as one or more zones and/or zone groups the playback device 200 is a part of, audio sources accessible by the playback device 200, or a playback queue that the playback device 200 (or some other playback device) may be associated with. The data may be stored as one or more state variables that are periodically updated and used to describe the state of the playback device 200. The memory 206 may also include the data associated with the state of the other devices of the media system, and shared from time to time among the devices so that one or more of the devices have the most recent data associated with the system. Other embodiments are also possible.

[0028] The audio processing components 208 may include one or more digital-to-analog converters (DAC), an audio preprocessing component, an audio enhancement component or a digital signal processor (DSP), and so on. In one embodiment, one or more of the audio processing components 208 may be a subcomponent of the processor 202. In one example, audio content may be processed and/or intentionally altered by the audio processing components 208 to produce audio signals. The produced audio signals may then be provided to the audio amplifier(s) 210 for amplification and playback through speaker(s) 212. Particularly, the audio amplifier(s) 210 may include devices configured to amplify audio signals to a level for driving one or more of the speakers 212. The speaker(s) 212 may include an individual transducer (e.g., a “driver”) or a complete speaker system involving an enclosure with one or more drivers. A particular driver of the speaker(s) 212 may include, for example, a subwoofer (e.g., for low frequencies), a mid-range driver (e.g., for middle frequencies), and/or a tweeter (e.g., for high frequencies). In some cases, each transducer in the one or more speakers 212 may be driven by an individual corresponding audio amplifier of the audio amplifier(s) 210. In addition to producing analog signals for playback by the playback device 200, the audio processing components 208 may be configured to process audio content to be sent to one or more other playback devices for playback.

[0029] Audio content to be processed and/or played back by the playback device 200 may be received from an external source, such as via an audio line-in input connection (e.g., an auto-detecting 3.5 mm audio line-in connection) or the network interface 214.

[0030] The network interface 214 may be configured to facilitate a data flow between the playback device 200 and one or more other devices on a data network. As such, the playback device 200 may be configured to receive audio content over the data network from one or more other playback devices in communication with the playback device 200, network devices within a local area network, or audio content sources over a wide area network such as the Internet. In one example, the audio content and other signals transmitted and received by the playback device 200 may be transmitted in the form of digital packet data containing an Internet Protocol (IP)-based source address and IP-based destination addresses. In such a case, the network interface 214 may be configured to parse the digital packet data such

that the data destined for the playback device 200 is properly received and processed by the playback device 200.

[0031] As shown, the network interface 214 may include wireless interface(s) 216 and wired interface(s) 218. The wireless interface(s) 216 may provide network interface functions for the playback device 200 to wirelessly communicate with other devices (e.g., other playback device(s), speaker(s), receiver(s), network device(s), control device(s) within a data network the playback device 200 is associated with) in accordance with a communication protocol (e.g., any wireless standard including IEEE 802.11a, 802.11b, 802.11g, 802.11n, 802.11ac, 802.15, 4G mobile communication standard, and so on). The wired interface(s) 218 may provide network interface functions for the playback device 200 to communicate over a wired connection with other devices in accordance with a communication protocol (e.g., IEEE 802.3). While the network interface 214 shown in FIG. 2 includes both wireless interface(s) 216 and wired interface(s) 218, the network interface 214 may in some embodiments include only wireless interface(s) or only wired interface(s).

[0032] In one example, the playback device 200 and one other playback device may be paired to play two separate audio components of audio content. For instance, playback device 200 may be configured to play a left channel audio component, while the other playback device may be configured to play a right channel audio component, thereby producing or enhancing a stereo effect of the audio content. The paired playback devices (also referred to as “bonded playback devices”) may further play audio content in synchrony with other playback devices.

[0033] In another example, the playback device 200 may be sonically consolidated with one or more other playback devices to form a single, consolidated playback device. A consolidated playback device may be configured to process and reproduce sound differently than an unconsolidated playback device or playback devices that are paired, because a consolidated playback device may have additional speaker drivers through which audio content may be rendered. For instance, if the playback device 200 is a playback device designed to render low frequency range audio content (i.e. a subwoofer), the playback device 200 may be consolidated with a playback device designed to render full frequency range audio content. In such a case, the full frequency range playback device, when consolidated with the low frequency playback device 200, may be configured to render only the mid and high frequency components of audio content, while the low frequency range playback device 200 renders the low frequency component of the audio content. The consolidated playback device may further be paired with a single playback device or yet another consolidated playback device.

[0034] By way of illustration, SONOS, Inc. presently offers (or has offered) for sale certain playback devices including a “PLAY:1,” “PLAY:3,” “PLAY:5,” “PLAYBAR,” “CONNECT:AMP,” “CONNECT,” and “SUB.” Any other past, present, and/or future playback devices may additionally or alternatively be used to implement the playback devices of example embodiments disclosed herein. Additionally, it is understood that a playback device is not limited to the example illustrated in FIG. 2 or to the SONOS product offerings. For example, a playback device may include a wired or wireless headphone. In another example, a playback device may include or interact with a docking station

for personal mobile media playback devices. In yet another example, a playback device may be integral to another device or component such as a television, a lighting fixture, or some other device for indoor or outdoor use.

b. Example Playback Zone Configurations

[0035] Referring back to the media playback system **100** of FIG. **1**, the environment may have one or more playback zones, each with one or more playback devices. The media playback system **100** may be established with one or more playback zones, after which one or more zones may be added, or removed to arrive at the example configuration shown in FIG. **1**. Each zone may be given a name according to a different room or space such as an office, bathroom, master bedroom, bedroom, kitchen, dining room, living room, and/or balcony. In one case, a single playback zone may include multiple rooms or spaces. In another case, a single room or space may include multiple playback zones.

[0036] As shown in FIG. **1**, the balcony, dining room, kitchen, bathroom, office, and bedroom zones each have one playback device, while the living room and master bedroom zones each have multiple playback devices. In the living room zone, playback devices **104**, **106**, **108**, and **110** may be configured to play audio content in synchrony as individual playback devices, as one or more bonded playback devices, as one or more consolidated playback devices, or any combination thereof. Similarly, in the case of the master bedroom, playback devices **122** and **124** may be configured to play audio content in synchrony as individual playback devices, as a bonded playback device, or as a consolidated playback device.

[0037] In one example, one or more playback zones in the environment of FIG. **1** may each be playing different audio content. For instance, the user may be grilling in the balcony zone and listening to hip hop music being played by the playback device **102** while another user may be preparing food in the kitchen zone and listening to classical music being played by the playback device **114**. In another example, a playback zone may play the same audio content in synchrony with another playback zone. For instance, the user may be in the office zone where the playback device **118** is playing the same rock music that is being playing by playback device **102** in the balcony zone. In such a case, playback devices **102** and **118** may be playing the rock music in synchrony such that the user may seamlessly (or at least substantially seamlessly) enjoy the audio content that is being played out-loud while moving between different playback zones. Synchronization among playback zones may be achieved in a manner similar to that of synchronization among playback devices, as described in previously referenced U.S. Pat. No. 8,234,395.

[0038] As suggested above, the zone configurations of the media playback system **100** may be dynamically modified, and in some embodiments, the media playback system **100** supports numerous configurations. For instance, if a user physically moves one or more playback devices to or from a zone, the media playback system **100** may be reconfigured to accommodate the change(s). For instance, if the user physically moves the playback device **102** from the balcony zone to the office zone, the office zone may now include both the playback device **118** and the playback device **102**. The playback device **102** may be paired or grouped with the office zone and/or renamed if so desired via a control device such as the control devices **126** and **128**. On the other hand, if the one or more playback devices are moved to a particular

area in the home environment that is not already a playback zone, a new playback zone may be created for the particular area.

[0039] Further, different playback zones of the media playback system **100** may be dynamically combined into zone groups or split up into individual playback zones. For instance, the dining room zone and the kitchen zone **114** may be combined into a zone group for a dinner party such that playback devices **112** and **114** may render audio content in synchrony. On the other hand, the living room zone may be split into a television zone including playback device **104**, and a listening zone including playback devices **106**, **108**, and **110**, if the user wishes to listen to music in the living room space while another user wishes to watch television.

c. Example Control Devices

[0040] FIG. **3** shows a functional block diagram of an example control device **300** that may be configured to be one or both of the control devices **126** and **128** of the media playback system **100**. As shown, the control device **300** may include a processor **302**, memory **304**, a network interface **306**, and a user interface **308**. In one example, the control device **300** may be a dedicated controller for the media playback system **100**. In another example, the control device **300** may be a network device on which media playback system controller application software may be installed, such as for example, an iPhone™ iPad™ or any other smart phone, tablet or network device (e.g., a networked computer such as a PC or Mac™).

[0041] The processor **302** may be configured to perform functions relevant to facilitating user access, control, and configuration of the media playback system **100**. The memory **304** may be configured to store instructions executable by the processor **302** to perform those functions. The memory **304** may also be configured to store the media playback system controller application software and other data associated with the media playback system **100** and the user.

[0042] In one example, the network interface **306** may be based on an industry standard (e.g., infrared, radio, wired standards including IEEE 802.3, wireless standards including IEEE 802.11a, 802.11b, 802.11g, 802.11n, 802.11ac, 802.15, 4G mobile communication standard, and so on). The network interface **306** may provide a means for the control device **300** to communicate with other devices in the media playback system **100**. In one example, data and information (e.g., such as a state variable) may be communicated between control device **300** and other devices via the network interface **306**. For instance, playback zone and zone group configurations in the media playback system **100** may be received by the control device **300** from a playback device or another network device, or transmitted by the control device **300** to another playback device or network device via the network interface **306**. In some cases, the other network device may be another control device.

[0043] Playback device control commands such as volume control and audio playback control may also be communicated from the control device **300** to a playback device via the network interface **306**. As suggested above, changes to configurations of the media playback system **100** may also be performed by a user using the control device **300**. The configuration changes may include adding/removing one or more playback devices to/from a zone, adding/removing one or more zones to/from a zone group, forming a bonded or consolidated player, separating one or more playback

devices from a bonded or consolidated player, among others. Accordingly, the control device 300 may sometimes be referred to as a controller, whether the control device 300 is a dedicated controller or a network device on which media playback system controller application software is installed.

[0044] The user interface 308 of the control device 300 may be configured to facilitate user access and control of the media playback system 100, by providing a controller interface such as the controller interface 400 shown in FIG. 4. The controller interface 400 includes a playback control region 410, a playback zone region 420, a playback status region 430, a playback queue region 440, and an audio content sources region 450. The user interface 400 as shown is just one example of a user interface that may be provided on a network device such as the control device 300 of FIG. 3 (and/or the control devices 126 and 128 of FIG. 1) and accessed by users to control a media playback system such as the media playback system 100. Other user interfaces of varying formats, styles, and interactive sequences may alternatively be implemented on one or more network devices to provide comparable control access to a media playback system.

[0045] The playback control region 410 may include selectable (e.g., by way of touch or by using a cursor) icons to cause playback devices in a selected playback zone or zone group to play or pause, fast forward, rewind, skip to next, skip to previous, enter/exit shuffle mode, enter/exit repeat mode, enter/exit cross fade mode. The playback control region 410 may also include selectable icons to modify equalization settings, and playback volume, among other possibilities.

[0046] The playback zone region 420 may include representations of playback zones within the media playback system 100. In some embodiments, the graphical representations of playback zones may be selectable to bring up additional selectable icons to manage or configure the playback zones in the media playback system, such as a creation of bonded zones, creation of zone groups, separation of zone groups, and renaming of zone groups, among other possibilities.

[0047] For example, as shown, a “group” icon may be provided within each of the graphical representations of playback zones. The “group” icon provided within a graphical representation of a particular zone may be selectable to bring up options to select one or more other zones in the media playback system to be grouped with the particular zone. Once grouped, playback devices in the zones that have been grouped with the particular zone will be configured to play audio content in synchrony with the playback device(s) in the particular zone. Analogously, a “group” icon may be provided within a graphical representation of a zone group. In this case, the “group” icon may be selectable to bring up options to deselect one or more zones in the zone group to be removed from the zone group. Other interactions and implementations for group and ungroup zones via a user interface such as the user interface 400 are also possible. The representations of playback zones in the playback zone region 420 may be dynamically updated as playback zone or zone group configurations are modified.

[0048] The playback status region 430 may include graphical representations of audio content that is presently being played, previously played, or scheduled to play next in the selected playback zone or zone group. The selected playback zone or zone group may be visually distinguished

on the user interface, such as within the playback zone region 420 and/or the playback status region 430. The graphical representations may include track title, artist name, album name, album year, track length, and other relevant information that may be useful for the user to know when controlling the media playback system via the user interface 400.

[0049] The playback queue region 440 may include graphical representations of audio content in a playback queue associated with the selected playback zone or zone group. In some embodiments, each playback zone or zone group may be associated with a playback queue containing information corresponding to zero or more audio items for playback by the playback zone or zone group. For instance, each audio item in the playback queue may comprise a uniform resource identifier (URI), a uniform resource locator (URL) or some other identifier that may be used by a playback device in the playback zone or zone group to find and/or retrieve the audio item from a local audio content source or a networked audio content source, possibly for playback by the playback device.

[0050] In one example, a playlist may be added to a playback queue, in which case information corresponding to each audio item in the playlist may be added to the playback queue. In another example, audio items in a playback queue may be saved as a playlist. In a further example, a playback queue may be empty, or populated but “not in use” when the playback zone or zone group is playing continuously streaming audio content, such as Internet radio that may continue to play until otherwise stopped, rather than discrete audio items that have playback durations. In an alternative embodiment, a playback queue can include Internet radio and/or other streaming audio content items and be “in use” when the playback zone or zone group is playing those items. Other examples are also possible.

[0051] When playback zones or zone groups are “grouped” or “ungrouped,” playback queues associated with the affected playback zones or zone groups may be cleared or re-associated. For example, if a first playback zone including a first playback queue is grouped with a second playback zone including a second playback queue, the established zone group may have an associated playback queue that is initially empty, that contains audio items from the first playback queue (such as if the second playback zone was added to the first playback zone), that contains audio items from the second playback queue (such as if the first playback zone was added to the second playback zone), or a combination of audio items from both the first and second playback queues. Subsequently, if the established zone group is ungrouped, the resulting first playback zone may be re-associated with the previous first playback queue, or be associated with a new playback queue that is empty or contains audio items from the playback queue associated with the established zone group before the established zone group was ungrouped. Similarly, the resulting second playback zone may be re-associated with the previous second playback queue, or be associated with a new playback queue that is empty, or contains audio items from the playback queue associated with the established zone group before the established zone group was ungrouped. Other examples are also possible.

[0052] Referring back to the user interface 400 of FIG. 4, the graphical representations of audio content in the playback queue region 440 may include track titles, artist names,

track lengths, and other relevant information associated with the audio content in the playback queue. In one example, graphical representations of audio content may be selectable to bring up additional selectable icons to manage and/or manipulate the playback queue and/or audio content represented in the playback queue. For instance, a represented audio content may be removed from the playback queue, moved to a different position within the playback queue, or selected to be played immediately, or after any currently playing audio content, among other possibilities. A playback queue associated with a playback zone or zone group may be stored in a memory on one or more playback devices in the playback zone or zone group, on a playback device that is not in the playback zone or zone group, and/or some other designated device.

[0053] The audio content sources region **450** may include graphical representations of selectable audio content sources from which audio content may be retrieved and played by the selected playback zone or zone group. Discussions pertaining to audio content sources may be found in the following section.

d. Example Audio Content Sources

[0054] As indicated previously, one or more playback devices in a zone or zone group may be configured to retrieve for playback audio content (e.g. according to a corresponding URI or URL for the audio content) from a variety of available audio content sources. In one example, audio content may be retrieved by a playback device directly from a corresponding audio content source (e.g., a line-in connection). In another example, audio content may be provided to a playback device over a network via one or more other playback devices or network devices.

[0055] Example audio content sources may include a memory of one or more playback devices in a media playback system such as the media playback system **100** of FIG. **1**, local music libraries on one or more network devices (such as a control device, a network-enabled personal computer, or a networked-attached storage (NAS), for example), streaming audio services providing audio content via the Internet (e.g., the cloud), or audio sources connected to the media playback system via a line-in input connection on a playback device or network device, among other possibilities.

[0056] In some embodiments, audio content sources may be regularly added or removed from a media playback system such as the media playback system **100** of FIG. **1**. In one example, an indexing of audio items may be performed whenever one or more audio content sources are added, removed or updated. Indexing of audio items may involve scanning for identifiable audio items in all folders/directory shared over a network accessible by playback devices in the media playback system, and generating or updating an audio content database containing metadata (e.g., title, artist, album, track length, among others) and other associated information, such as a URI or URL for each identifiable audio item found. Other examples for managing and maintaining audio content sources may also be possible.

[0057] The above discussions relating to playback devices, controller devices, playback zone configurations, and media content sources provide only some examples of operating environments within which functions and methods described below may be implemented. Other operating environments and configurations of media playback systems, playback devices, and network devices not explicitly

described herein may also be applicable and suitable for implementation of the functions and methods.

III. Example Methods

[0058] As discussed above, embodiments described herein involve, among other things, providing access to and control of a media playback network via a media playback application. The embodiments further involve providing the capability to form playback groups of zone players, from within the media playback application or from a user interface that is accessible from the media playback application. And the embodiments further involve providing for the control of playback volumes associated with the zone players, from within the media playback application or from a user interface accessible from the media playback application. In some embodiments, such control of playback volumes of zone players within a playback group may be controlled before the playback group is formed. Other aspects of the embodiments will be made apparent in the remainder of description herein.

[0059] Method **500** shown in FIG. **5** presents an embodiment of a method that can be implemented within an operating environment involving, for example, the media playback system **100** of FIG. **1**, one or more of the playback device **200** of FIG. **2**, and one or more of the control device **300** of FIG. **3**. Method **500** may include one or more operations, functions, or actions as illustrated by one or more of blocks **502-506**. Although the blocks are illustrated in sequential order, these blocks may also be performed in parallel, and/or in a different order than those described herein. Also, the various blocks may be combined into fewer blocks, divided into additional blocks, and/or removed based upon the desired implementation.

[0060] In addition, for the method **500** and other processes and methods disclosed herein, the flowchart shows functionality and operation of one possible implementation of present embodiments. In this regard, each block may represent a module, a segment, or a portion of program code, which includes one or more instructions executable by a processor for implementing specific logical functions or steps in the process. The program code may be stored on any type of computer readable medium, for example, such as a storage device including a disk or hard drive. The computer readable medium may include non-transitory computer readable medium, for example, such as computer-readable media that stores data for short periods of time like register memory, processor cache and Random Access Memory (RAM). The computer readable medium may also include non-transitory media, such as secondary or persistent long term storage, like read only memory (ROM), optical or magnetic disks, compact-disc read only memory (CD-ROM), for example. The computer readable media may also be any other volatile or non-volatile storage systems. The computer readable medium may be considered a computer readable storage medium, for example, or a tangible storage device. In addition, for the method **500** and other processes and methods disclosed herein, each block in FIG. **5** may represent circuitry that is wired to perform the specific logical functions in the process.

[0061] It should be understood that, before initiating method **500** (i.e., commencing performance of block **502**), a device may cause a user interface to display a user interface having features such as that depicted in FIG. **7**. FIG. **7** may be understood to represent a display associated with a media

playback application. Such a media player application may be associated with a given playback device or network of playback devices, or may be independent of such a playback device or network of playback devices (i.e., it may be a “third party” application). The display such as that shown in FIG. 7 may include a number of features such as an indication of a media item, playback controls, and other features. The display may also include an activation icon 702. Upon a user touching or clicking the activation icon 702, the device may enter a state in which the device may display any of the displays of FIGS. 8-12, among other example displays. The user interface depicted in FIG. 7 may include other features and/or be arranged to perform other functions as well.

[0062] It should be noted that any of the displays depicted in FIGS. 8-12 can be displayed by a user interface as part of, or otherwise integrated into, a third-party media playback application, or may be part of a separate application that is launched upon the user clicking or touching the activation icon 702. Also, after navigating the displays represented by any of FIGS. 8-12, the controller device may revert back to displaying a display in accordance with the media playback application (e.g. the display of FIG. 7, or a similar display). The revision to the media playback application may occur automatically after completing operations described herein with regard to FIGS. 8-12, or may occur based on user input (e.g. a back command). Other examples are possible.

[0063] FIG. 8 depicts a display of multiple groups of zone players. For example, a first group may include a zone player associated with or located in a living room. A second group may include zone players associated with or located in a dining room, a kitchen, a master bedroom, and a guest bedroom. A third group may include zone players associated with or located in a patio and a foyer. A fourth group may include a zone player associated with or located in an entrance. Other such example groups may exist.

[0064] A user selecting a segment of the display associated with the living room group may cause a volume indication associated with the living room group to appear, as shown in FIG. 9. In turn, a user may touch and drag a volume slider 902 to increase or decrease the volume associated with the living room group of zone players.

[0065] At block 502 (i), method 500 includes providing a user interface that displays an indication of a respective volume corresponding to each of at least two zone players. The zone player of block 502 (i) may be similar to any playback device described herein.

[0066] In accordance with block 502 (i), FIG. 11 depicts example indications of respective volumes corresponding to three zone players including a kitchen zone, a patio zone, and a living room zone. Volume indication 1102 includes volume slider 1102A and volume indicator bar 1102B. Volume slider 1102A and volume indicator bar 1102B function together to indicate a relative volume of a zone player associated with a kitchen playback zone. For example, if volume indicator bar 1102B indicates a potential range of volumes ranging from 0-100%, volume slider 1102A lies upon volume indicator bar 1102B at a position that indicates the current volume of the kitchen playback zone is approximately 75%.

[0067] The volume indications 1104 and 1106, corresponding to a patio playback zone and a living room playback zone may respectively, indicate volumes of approximately 30% and 80%. FIG. 11 includes a zone player

identifier 1108, which may indicate a room that a corresponding zone player is located in or otherwise associated with. FIG. 12 includes similar example volume indications 1222 and 1232 that may function in a manner similar to volume indications 1102 and 1104 of FIG. 11.

[0068] At block 502 (ii) the method includes providing a user interface that displays a respective selectable icon corresponding to each of the at least two zone players. Each respective selectable icon indicates whether the corresponding zone player has been selected for a group and at least one of the respective selectable icons indicates that a respective zone player has been selected for the group.

[0069] For example, FIG. 10 depicts example selectable icons corresponding to eight zone players. By way of illustration, icon 1002 corresponds to a dining room zone player and icon 1004 corresponds to a kitchen zone player. Icon 1002 indicates that the dining room zone player has not been selected for a group, while icon 1004 indicates that the kitchen zone player has been selected for the group. The display of FIG. 10 may appear on a touchscreen of a controller device, enabling a user to toggle or select any of the icons. For example, if a user were to touch the icon 1002, indicating a selection of the dining room zone player for inclusion in the group, the icon 1002 may change states to appear “checked,” similar to the appearance of icon 1004 in FIG. 10. Similarly, if the user were to click or touch icon 1004, the icon 1004 may change states to “unchecked” (i.e. deselected) and may indicate that the kitchen zone player is no longer included in the group. (For example, if the icon 1004 is deselected, adjustments to the group volume may leave the volume of the kitchen zone player unaffected.) FIG. 12 includes similar selectable icons 1202, 1204, and 1206 that may function in a similar manner.

[0070] At block 502 (iii) the method includes providing a user interface that displays an indication of a group volume associated with the at least one selected zone player. FIG. 11 depicts an example indication of a group volume. Group volume indication 1112 includes group volume slider 1112A and group volume indicator bar 1112B. Group volume slider 1112A and group volume indicator bar 1112B function together to indicate a relative group volume of selected zone players. For example, if group volume indicator bar 1112B indicates a potential range of volumes ranging from 0-100%, group volume slider 1112A lies upon volume indicator bar 1112B at a position that indicates the current group volume of the selected playback zones is approximately 40%. FIG. 12 includes a similar group volume indication 1212 that may function in a similar manner.

[0071] In one embodiment, the indications of volumes of zone players and the selectable icons (i.e., the displays discussed above with respect to block 502(i-iii)) may be displayed simultaneously. For example, FIG. 12 includes group volume indication 1212, volume indications 1222 and 1232, and also selectable icons 1202, 1204, and 1206. Selectable icons 1202, 1204, and 1206 indicate that the zone player respectively corresponding to the kitchen, patio, and living room are selected for inclusion in the group while the volumes of the respective zone players are simultaneously indicated by the volume indications. By way of illustration, selectable icon 1208 indicates that a zone player associated with an office is not included in the group.

[0072] At block 504, the method includes receiving a command to change at least one of (i) a respective volume of one or more of the at least one selected zone player or (ii)

the group volume. For example, a user may be able to manipulate a touchscreen of a controller device to change a volume of a selected zone player or to change a group volume. Referring to FIG. 11, the user may touch volume slider 1102A and drag it to the left, causing a reduction in a volume associated with the kitchen zone player. Likewise, dragging volume slider 1102A to the right may increase the volume of the kitchen zone player. Also, the user may touch and drag group volume slider 1112A to the left or to the right, to cause a respective increase or decrease in the group volume. In other embodiments, the user may change volumes by clicking and dragging the volume sliders via a mouse. Other examples are possible.

[0073] Prior to receiving the command in accordance with block 504, the controller device may receive a selection of the at least one selected zone player. In the example shown in FIG. 11, selections may have been received corresponding to the kitchen, patio, and living room zone players.

[0074] FIG. 12 may include volume sliders 1222A and 1232A that function in a manner similar to 1102A and 1104A.

[0075] At block 506, the method includes transmitting an indication of the received command to one or more of the at least one selected zone player. Referring to the example of FIG. 11, transmitting the indication of the received command may occur in response to a user clicking or touching the “Done” button of the user interface. The controller device may transmit the command to the zone players that are affected by the command received at block 504. For example, if, prior to touching the “done” button, a user drags volume slider 1102A and a volume slider 1104A, indications of those received commands will be transmitted, by the controller device, to the kitchen and patio zone players, respectively. Likewise, if the user drags the group volume slider 1112A prior to clicking the “done” button, respective volumes corresponding to each of the kitchen, patio, and living room zone players may be increased or decreased proportionally according to the received command. FIG. 12 includes a “Done” button that may function in a similar manner, causing transmission of commands pertaining to selected zone players. Other examples of causing the indication of the received command to be transmitted may exist.

[0076] The method may further include displaying the indication of the respective volume corresponding to each of the at least two zone players prior to displaying the respective selectable icons corresponding to each of the at least two zone players. For example, if the user interface is in a state such that it is displaying the arrangement of FIG. 11, the user may click the “Rooms” button to cause the user interface to display the arrangement of FIG. 10. In another embodiment, the selectable icons and the indications of volume corresponding to the at least two zone players may be displayed simultaneously, such as in FIG. 12.

[0077] The method may further include receiving a command to form the group with two or more of the at least two zone players and transmitting a command, to the two or more zone players, to form the group. In some embodiments, the user clicking or touching the “Done” button of FIG. 10-12 may indicate to form a group with the selected zone players and for the group to synchronously playback media content. This may occur upon the controller transmitting the command to form the group to the at least two zone players.

[0078] Method 600 shown in FIG. 6 presents an embodiment of a method that can be implemented within an

operating environment involving, for example, the media playback system 100 of FIG. 1, one or more of the playback device 200 of FIG. 2, and one or more of the control device 300 of FIG. 3. Method 600 may include one or more operations, functions, or actions as illustrated by one or more of blocks 602-608. Although the blocks are illustrated in sequential order, these blocks may also be performed in parallel, and/or in a different order than those described herein. Also, the various blocks may be combined into fewer blocks, divided into additional blocks, and/or removed based upon the desired implementation.

[0079] In addition, for the method 600 and other processes and methods disclosed herein, the flowchart shows functionality and operation of one possible implementation of present embodiments. In this regard, each block may represent a module, a segment, or a portion of program code, which includes one or more instructions executable by a processor for implementing specific logical functions or steps in the process. The program code may be stored on any type of computer readable medium, for example, such as a storage device including a disk or hard drive. The computer readable medium may include non-transitory computer readable medium, for example, such as computer-readable media that stores data for short periods of time like register memory, processor cache and Random Access Memory (RAM). The computer readable medium may also include non-transitory media, such as secondary or persistent long term storage, like read only memory (ROM), optical or magnetic disks, compact-disc read only memory (CD-ROM), for example. The computer readable media may also be any other volatile or non-volatile storage systems. The computer readable medium may be considered a computer readable storage medium, for example, or a tangible storage device. In addition, for the method 600 and other processes and methods disclosed herein, each block in FIG. 6 may represent circuitry that is wired to perform the specific logical functions in the process.

[0080] At block 602, method 600 may include receiving a selection of at least two zone players to be included in a group. For example, referring to FIG. 12, a user may click or touch selectable icons such as icons 1202 or 1204 to select or deselect zone players associated with a kitchen and a patio, respectively.

[0081] At block 604, the method may include causing a graphical interface to display (i) an indication of a respective volume of each of the at least two selected zone players and (ii) an indication of a group volume associated with the at least two selected zone players. For example, a controller device may display an arrangement such as that depicted in FIG. 12. FIG. 12 includes volume indications 1222 and 1232, and group volume indication 1212.

[0082] At optional block 606, the method may include, subsequent to causing the graphical interface to display the indications, receiving a command to form the group of the at least two selected zone players. For example, a user may command that the group be formed by clicking or touching the “Done” button of FIG. 12. In this case, a group including zone players associated with the kitchen, the patio, and the living room would be formed as the group.

[0083] At optional block 608, the method may include transmitting a command to each of the at least two selected zone players to synchronously play a portion of media content. This may occur in response to receiving the command to form the group. That is, groups of zone players may

be configured to collectively play back a portion of media content synchronously. Synchronous play of the portion of media content may include multiple zone players playing a common portion of media content so that there is no delay or lag between the zone players.

[0084] The method may also include forming the group of the at least two zone players. For example, the zone players that make up the group may be caused to sync and collectively play a common portion of media content for synchronous playback.

[0085] The method may also include receiving a command to change a respective volume of at least one of the at least two selected zone players. For example, the user may touch and drag volume slider 1232A, indicating to change the volume corresponding to the patio zone player, perhaps before, or perhaps contemporaneous with, providing the command to form the group or to synchronously play the portion of media content. The command to change the volume of at least one of the at least two selected zone players may be transmitted to the respective zone players of which the user wishes to change volumes.

IV. Conclusion

[0086] The description above discloses, among other things, various example systems, methods, apparatus, and articles of manufacture including, among other components, firmware and/or software executed on hardware. It is understood that such examples are merely illustrative and should not be considered as limiting. For example, it is contemplated that any or all of the firmware, hardware, and/or software aspects or components can be embodied exclusively in hardware, exclusively in software, exclusively in firmware, or in any combination of hardware, software, and/or firmware. Accordingly, the examples provided are not the only way(s) to implement such systems, methods, apparatus, and/or articles of manufacture.

[0087] In one aspect, a method is provided. The method involves providing a user interface that simultaneously (i) displays an indication of a respective volume corresponding to each of at least two zone players and (ii) displays a respective selectable icon corresponding to each of the at least two zone players. Each respective selectable icon indicates whether the corresponding zone player has been selected for a group, and at least one of the respective selectable icons indicates that a respective zone player has been selected for the group. The interface further simultaneously displays (iii) an indication of a group volume associated with the at least one selected zone player. The method further includes receiving a command to change at least one of (i) a respective volume of one or more of the at least one selected zone player or (ii) the group volume, and transmitting an indication of the received command to one or more of the at least one selected zone player.

[0088] In another aspect, a device is provided. The device includes a processor and memory with instructions stored therein, that when executed cause the device to perform functions. The functions comprise providing a user interface that simultaneously (i) displays an indication of a respective volume corresponding to each of at least two zone players and (ii) displays a respective selectable icon corresponding to each of the at least two zone players. Each respective selectable icon indicates whether the corresponding zone player has been selected for a group, and at least one of the respective selectable icons indicates that a respective zone

player has been selected for the group. The interface further simultaneously displays (iii) an indication of a group volume associated with the at least one selected zone player. The functions further comprise receiving a command to change at least one (i) of a respective volume of one or more of the at least one selected zone player or (ii) the group volume. The functions further comprise transmitting an indication of the received command to one or more of the at least one selected zone player.

[0089] In yet another aspect, a non-transitory computer readable memory is provided. The non-transitory computer readable memory has stored thereon instructions executable by a computing device to cause the computing device to perform functions. The functions comprise providing a user interface that simultaneously (i) displays an indication of a respective volume corresponding to each of at least two zone players and (ii) displays a respective selectable icon corresponding to each of the at least two zone players. Each respective selectable icon indicates whether the corresponding zone player has been selected for a group, and at least one of the respective selectable icons indicates that a respective zone player has been selected for the group. The interface further simultaneously displays (iii) an indication of a group volume associated with the at least one selected zone player. The functions further comprise receiving a command to change at least one of (i) a respective volume of one or more of the at least one selected zone player or (ii) the group volume. The functions further comprise transmitting an indication of the received command to one or more of the at least one selected zone player.

[0090] In yet another aspect, a method is provided. The method involves receiving a selection of at least two zone players to be included in a group and causing a graphical interface to display (i) an indication of a respective volume of each of the at least two selected zone players and (ii) an indication of a group volume associated with the at least two selected zone players. The method further involves, subsequent to causing the graphical interface to display the indications, receiving a command to form the group of the at least two selected zone players; and transmitting a command to each of the at least two selected zone players to synchronously play a portion of media content.

[0091] In another aspect, a device is provided. The device includes a processor and memory with instructions stored therein, that when executed cause the device to perform functions. The functions comprise receiving a selection of at least two zone players to be included in a group and causing a graphical interface to display (i) an indication of a respective volume of each of the at least two selected zone players and (ii) an indication of a group volume associated with the at least two selected zone players. The functions further comprise, subsequent to causing the graphical interface to display the indications, receiving a command to form the group of the at least two selected zone players; and transmitting a command to each of the at least two selected zone players to synchronously play a portion of media content.

[0092] In yet another aspect, a non-transitory computer readable memory is provided. The non-transitory computer readable memory has stored thereon instructions executable by a computing device to cause the computing device to perform functions. The functions comprise receiving a selection of at least two zone players to be included in a group and causing a graphical interface to display (i) an indication of a respective volume of each of the at least two

selected zone players and (ii) an indication of a group volume associated with the at least two selected zone players. The functions further comprise, subsequent to causing the graphical interface to display the indications, receiving a command to form the group of the at least two selected zone players; and transmitting a command to each of the at least two selected zone players to synchronously play a portion of media content.

[0093] Additionally, references herein to “embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment can be included in at least one example embodiment of an invention. The appearances of this phrase in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments mutually exclusive of other embodiments. As such, the embodiments described herein, explicitly and implicitly understood by one skilled in the art, can be combined with other embodiments.

[0094] The specification is presented largely in terms of illustrative environments, systems, procedures, steps, logic blocks, processing, and other symbolic representations that directly or indirectly resemble the operations of data processing devices coupled to networks. These process descriptions and representations are typically used by those skilled in the art to most effectively convey the substance of their work to others skilled in the art. Numerous specific details are set forth to provide a thorough understanding of the present disclosure. However, it is understood to those skilled in the art that certain embodiments of the present disclosure can be practiced without certain, specific details. In other instances, well known methods, procedures, components, and circuitry have not been described in detail to avoid unnecessarily obscuring aspects of the embodiments. Accordingly, the scope of the present disclosure is defined by the appended claims rather than the forgoing description of embodiments.

[0095] When any of the appended claims are read to cover a purely software and/or firmware implementation, at least one of the elements in at least one example is hereby expressly defined to include a tangible, non-transitory medium such as a memory, DVD, CD, Blu-ray, and so on, storing the software and/or firmware.

1. A non-transitory computer-readable medium having stored thereon instructions executable by one or more processors to cause a computing device to perform functions comprising:

in response to receiving a first user command associated with grouping a first playback device with a second playback device in a synchrony group while the second playback device is already playing media content from a playlist, causing the first playback device to play a portion of the media content in synchrony with the second playback device;

displaying in a graphical user interface (GUI), (i) a first volume icon at a first position indicating a volume level of the first playback device, (ii) a second volume icon at a second position indicating a volume level of the second playback device, and (iii) a group volume icon at a third position indicating a group volume level of the synchrony group;

in response to receiving a second user command associated with adjusting the volume level of the first playback device, (i) adjusting the first volume icon to a

fourth position that corresponds to the second user command, and (ii) causing the volume level of the first playback device to be adjusted to a level corresponding to the second user command;

in response to receiving a third user command associated with adjusting the volume level of the second playback device, (i) adjusting the second volume icon to a fifth position that corresponds to the third user command, and (ii) causing the volume level of the second playback device to be adjusted to a level corresponding to the third user command; and

in response to receiving a fourth user command associated with adjusting the group volume level of the synchrony group, (i) adjusting the group volume icon to a sixth position that corresponds to the fourth user command, (ii) adjusting the first volume icon to a seventh position that is proportional to the fourth user command, (iii) adjusting the second volume icon to an eighth position that is proportional to the fourth user command, (iv) causing the volume level of the first playback device to be adjusted to a level that is proportional to the fourth user command, and (v) causing the volume level of the second playback device to be adjusted to a level that is proportional to the fourth user command.

2. The non-transitory computer-readable medium of claim 1, further comprising:

in response to receiving the second user command associated with adjusting the volume level of the first playback device, continuing to display in the GUI the second volume icon at the second position.

3. The non-transitory computer-readable medium of claim 1, further comprising:

in response to receiving the third user command associated with adjusting the volume level of the second playback device, continuing to display in the GUI the first volume icon at the first position.

4. The non-transitory computer-readable medium of claim 1, further comprising:

in response to receiving the first user command associated with grouping the first playback device with the second playback device in the synchrony group, displaying in the GUI (i) an indication that the first playback device is grouped in the synchrony group, and (ii) an indication that the second playback device is grouped in the synchrony group.

5. The non-transitory computer-readable medium of claim 4, wherein the GUI further comprises an indication that one or more additional playback devices are not grouped in the synchrony group.

6. The non-transitory computer-readable medium of claim 1, wherein the first position indicates a volume level that is greater than the third position.

7. The non-transitory computer-readable medium of claim 6, wherein the second position indicates a volume level that is also greater than the third position.

8. The non-transitory computer-readable medium of claim 1, wherein prior to receiving the first user command associated with grouping the first playback device with the second playback device in the synchrony group, the second playback device is already in the synchrony group with a third playback device, and wherein the first user command is received while the second playback device is already playing the media content from the playlist in synchrony with the third playback device.

9. The non-transitory computer-readable medium of claim 1, wherein the first volume icon comprises (i) an indicator bar, and (ii) a moveable indicator positioned relative to the indicator bar to indicate the volume level of the first playback device.

10. The non-transitory computer-readable medium of claim 1, wherein the media content comprises a song.

11. A method comprising:

in response to receiving a first user command associated with grouping a first playback device with a second playback device in a synchrony group while the second playback device is already playing media content from a playlist, causing the first playback device to play a portion of the media content in synchrony with the second playback device;

displaying in a graphical user interface (GUI), (i) a first volume icon at a first position indicating a volume level of the first playback device, (ii) a second volume icon at a second position indicating a volume level of the second playback device, and (iii) a group volume icon at a third position indicating a group volume level of the synchrony group;

in response to receiving a second user command associated with adjusting the volume level of the first playback device, (i) adjusting the first volume icon to a fourth position that corresponds to the second user command, and (ii) causing the volume level of the first playback device to be adjusted to a level corresponding to the second user command;

in response to receiving a third user command associated with adjusting the volume level of the second playback device, (i) adjusting the second volume icon to a fifth position that corresponds to the third user command, and (ii) causing the volume level of the second playback device to be adjusted to a level corresponding to the third user command; and

in response to receiving a fourth user command associated with adjusting the group volume level of the synchrony group, (i) adjusting the group volume icon to a sixth position that corresponds to the fourth user command, (ii) adjusting the first volume icon to a seventh position that is proportional to the fourth user command, (iii) adjusting the second volume icon to an eighth position that is proportional to the fourth user command, (iv) causing the volume level of the first playback device to be adjusted to a level that is proportional to the fourth

user command, and (v) causing the volume level of the second playback device to be adjusted to a level that is proportional to the fourth user command.

12. The method of claim 11, further comprising:

in response to receiving the second user command associated with adjusting the volume level of the first playback device, continuing to display in the GUI the second volume icon at the second position.

13. The method of claim 11, further comprising:

in response to receiving the third user command associated with adjusting the volume level of the second playback device, continuing to display in the GUI the first volume icon at the first position.

14. The method of claim 11, further comprising:

in response to receiving the first user command associated with grouping the first playback device with the second playback device in the synchrony group, displaying in the GUI (i) an indication that the first playback device is grouped in the synchrony group, and (ii) an indication that the second playback device is grouped in the synchrony group.

15. The method of claim 14, wherein the GUI further comprises an indication that one or more additional playback devices are not grouped in the synchrony group.

16. The method of claim 11, wherein the first position indicates a volume level that is greater than the third position.

17. The method of claim 16, wherein the second position indicates a volume level that is also greater than the third position.

18. The method of claim 11, wherein prior to receiving the first user command associated with grouping the first playback device with the second playback device in the synchrony group, the second playback device is already in the synchrony group with a third playback device, and wherein the first user command is received while the second playback device is already playing the media content from the playlist in synchrony with the third playback device.

19. The method of claim 11, wherein the first volume icon comprises (i) an indicator bar, and (ii) a moveable indicator positioned relative to the indicator bar to indicate the volume level of the first playback device.

20. The method of claim 11, wherein the media content comprises a song.

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